Exploring Classic R Datasets

Russ Blessing

2026-04-27

A short reproducible analysis of the iris and mtcars datasets, demonstrating how figures generated in a separate analysis notebook can be embedded directly into a Quarto manuscript. All figures are produced with ggplot2 and sourced from notebooks/analysis.qmd.

Introduction

This manuscript demonstrates a reproducible workflow using Quarto. All figures are generated in a dedicated analysis notebook (`notebooks/analysis.qmd`) and embedded here using the `{{< embed >}}` shortcode. Updating the notebook automatically updates this manuscript — no copy-paste required.

Iris Dataset

The `iris` dataset contains measurements of sepal and petal dimensions for 150 flowers across three species: *setosa*, *versicolor*, and *virginica* (Anderson 1935).

Sepal Dimensions

Figure 1 shows a clear separation between species in sepal space, with *setosa* occupying a distinct region of high width relative to length.

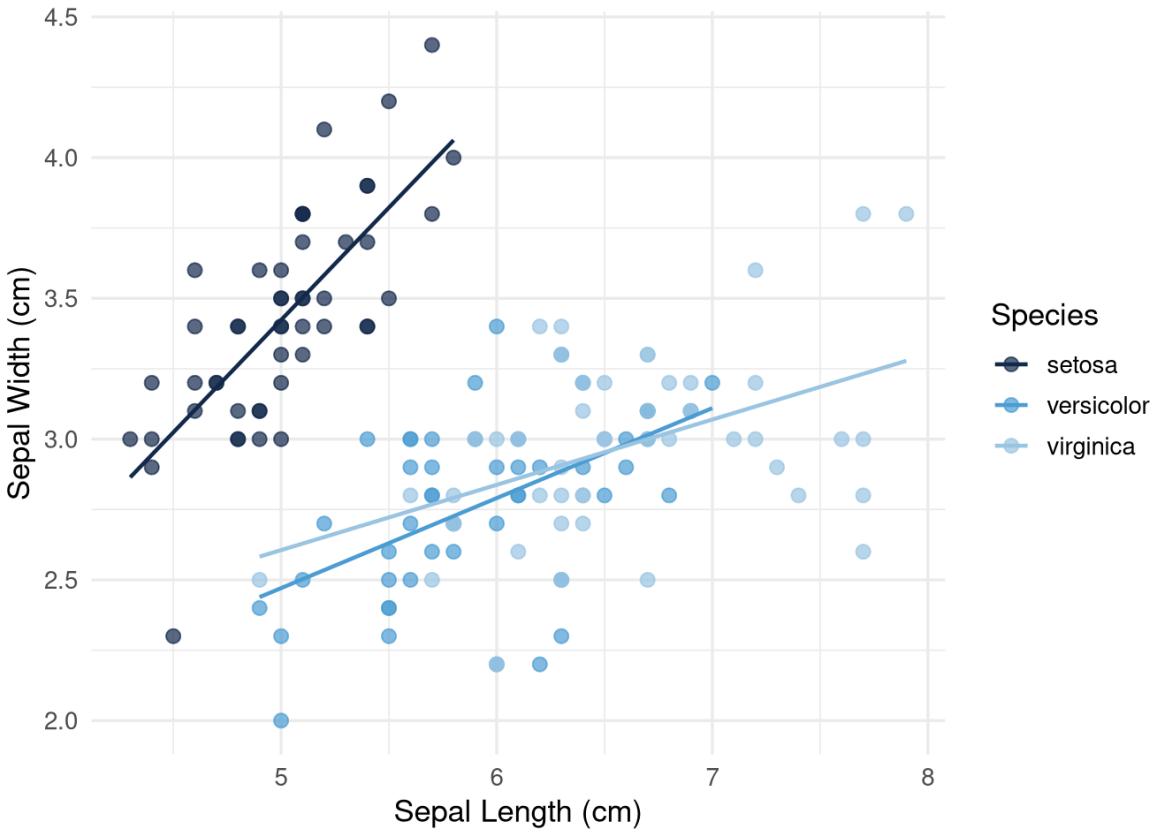

Figure 1: Sepal length vs. width across three iris species

Source: Analysis Notebook

Petal Length

Species separation is even more pronounced in petal length (Figure 2), making it a reliable discriminating variable.

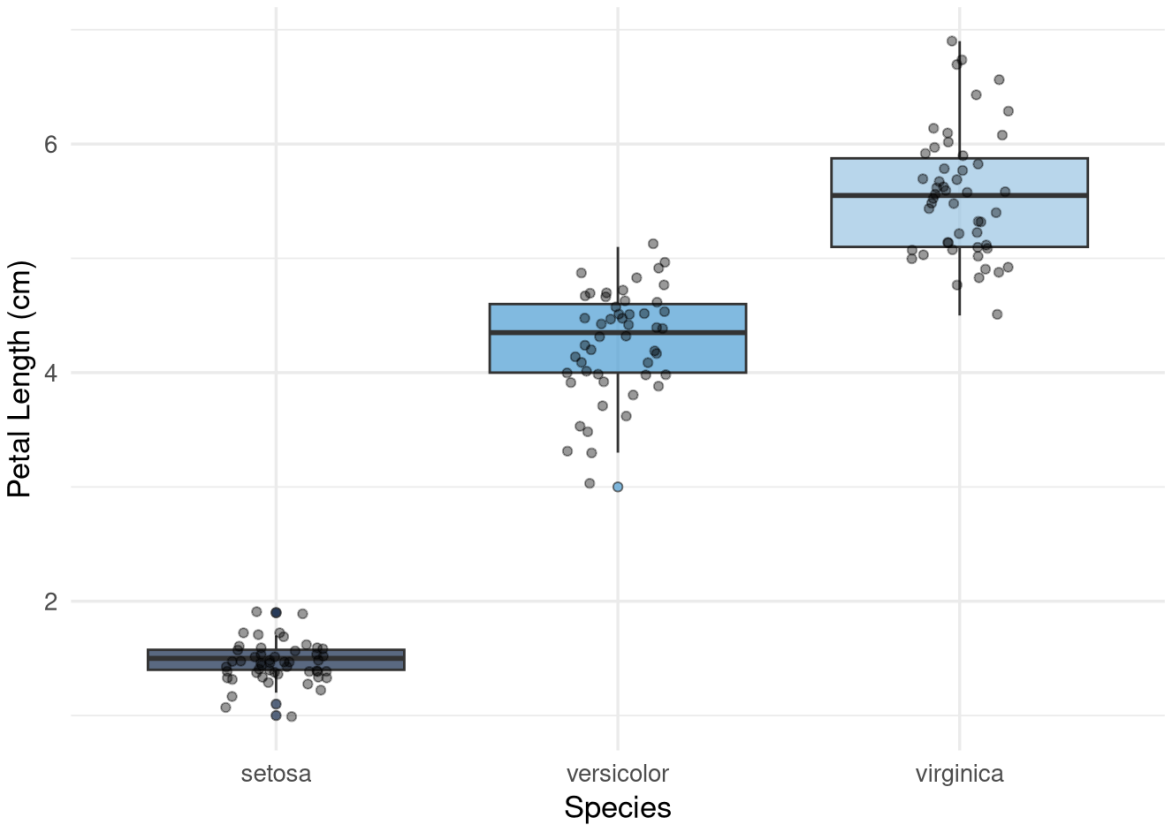

Figure 2: Distribution of petal length by species

Source: Analysis Notebook

Motor Trend Cars

The `mtcars` dataset records fuel efficiency and physical characteristics for 32 automobiles from the 1974 Motor Trend magazine (Henderson and Velleman 1981). Figure 3 illustrates the well-known negative relationship between vehicle weight and fuel efficiency, stratified by cylinder count.

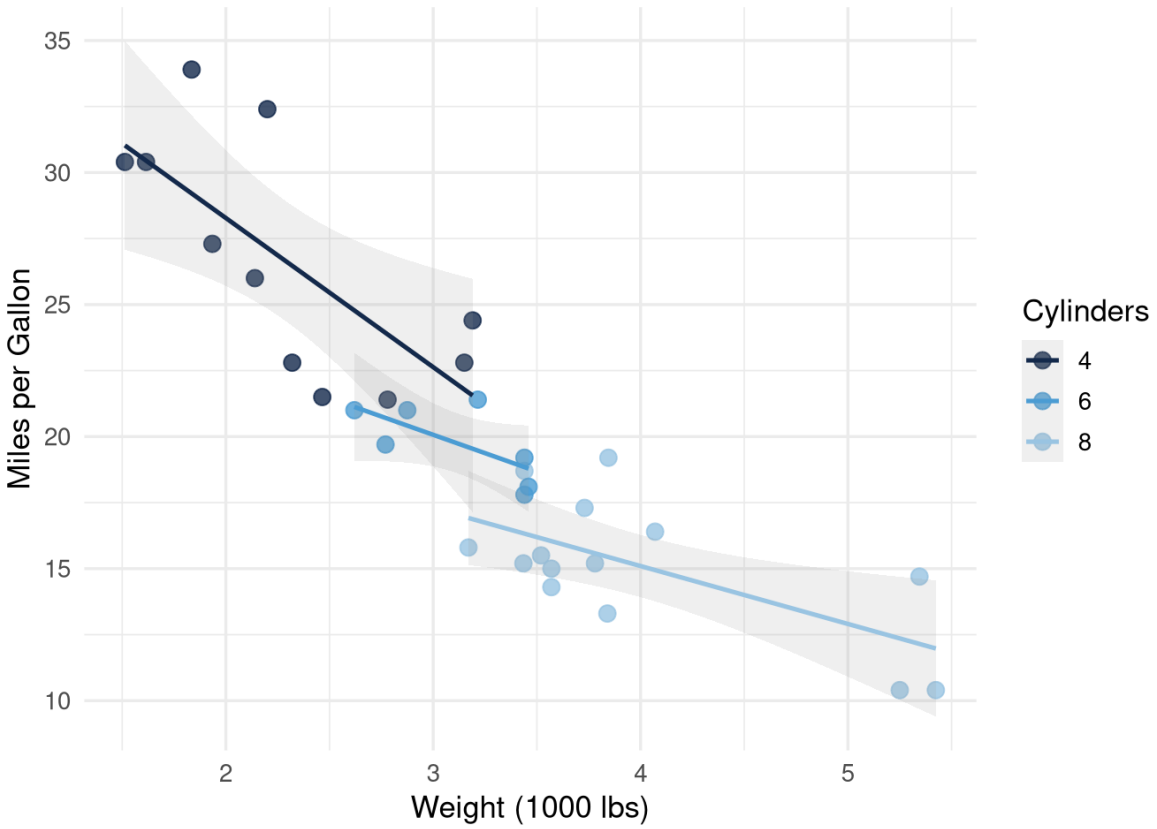

Figure 3: Vehicle weight vs. fuel efficiency, by number of cylinders

Source: Analysis Notebook

Conclusion

This document was rendered entirely from plain-text source files. The figures above were produced in `notebooks/analysis.qmd` and embedded without manual copying. Any change to the analysis notebook is reflected here on the next render.

References

Anderson, Edgar. 1935. "The Irises of the Gaspe Peninsula." *Bulletin of the American Iris Society* 59: 2–5.

Henderson, Harold V, and Paul F Velleman. 1981. "Building Multiple Regression Models Interactively." *Biometrics* 37 (2): 391–411.